

Name: _____

Date: _____

Determining Launch Parameters To Hit Targets

10% bonus credit — Module 2 exam

SYNOPSIS

Your task is to determine launch parameters — launch velocity and launch angle(s) — to hit targets at specified horizontal distances from the launcher on my bench table.

OBJECTIVES

Determine launch angles to hit targets at horizontal distances 1 m, 2 m, and 3 m, launched from my bench table.

Instructions

1. The first task is to find the launch velocity of the launcher. Describe a method to determine the launch velocity and apply the method to find the launch velocity. The launcher is fitted with a photogate that can independently verify the launch velocity. Compare the velocities from the two measurements. Which do you think is more reliable and why?

2. Calculate the launch angles using the launch velocity, launch height, and distances to the target.

Determining Launch Parameters To Hit Targets (10% bonus credit for Module 2 Exam)

Synopsis: In this activity, your task is to determine launch parameters, such as launch velocity and launch angles, to hit targets at specified horizontal distances away from the launcher.

Objectives: Determine launch angles to hit targets at horizontal distances, 1 m, 2 m, and 3 m, launched from my bench table.

Instructions

1. The first task is to find the launch velocity of the launcher. Describe a method to determine the launch velocity and apply the method to find the launch velocity. The launcher is fitted with a photogate that can independently verify the launch velocity. Compare the velocities from the two measurements. Which do you think is more reliable and why?

2. Calculate the launch angles using the launch velocity, launch height, and distances to the target.

Constant launch parameters

Launch velocity:

Launch height:

Show the calculation below.

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Original worksheet page (Dr. Cho) — use the space below to organize your own values and the calculation.

Constant launch parameters

Launch velocity (timed vertical launch): _____ m/s

Launch velocity (photogate): _____ m/s

Which is more reliable, and why?

Launch height, H : _____ m

Show the calculation below.

Launch angles to hit each target

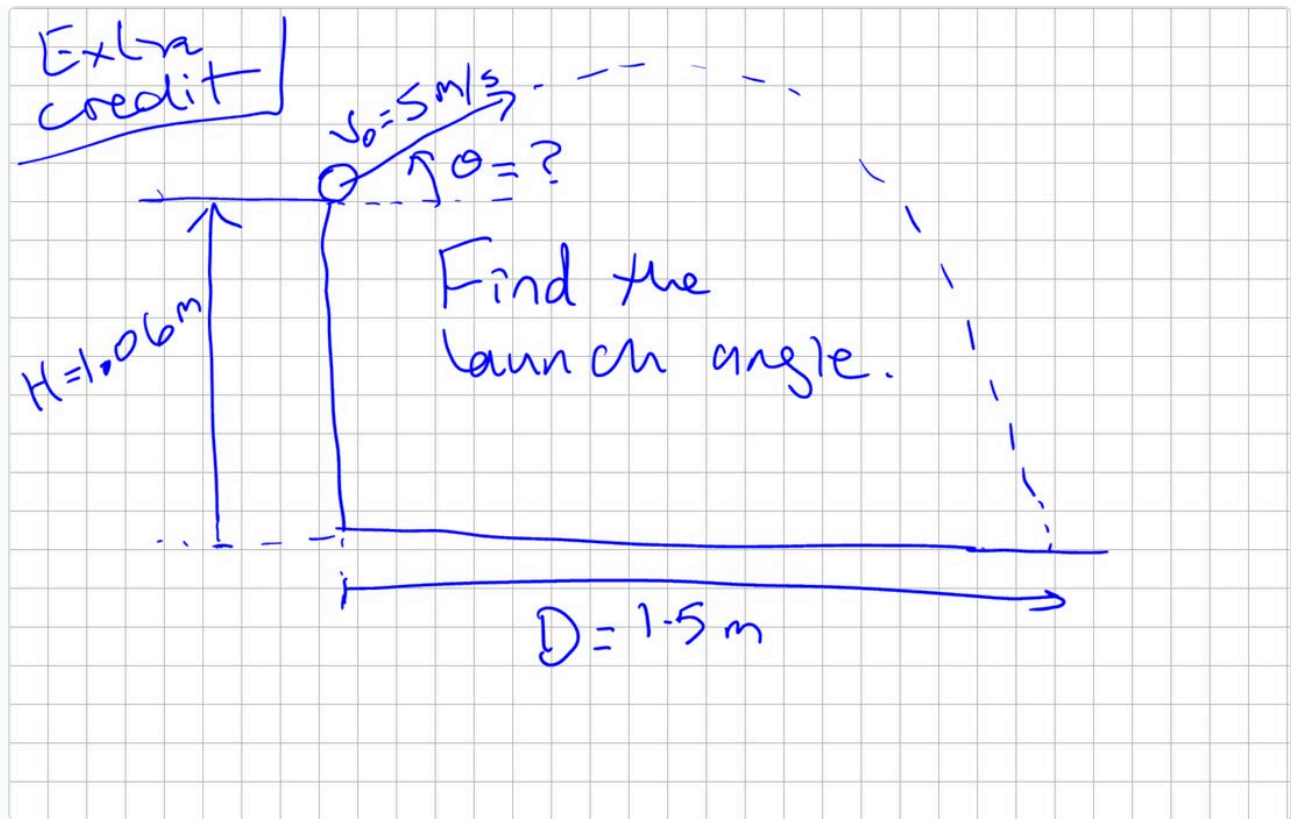
For each target distance, a launch angle problem generally has **two** solutions — a high arc and a low arc. Record both, then note which one you actually used at the bench and why.

TARGET DISTANCE, D	HIGH-ARC θ_1	LOW-ARC θ_2	ANGLE USED & WHY
1 m			
2 m			
3 m			

Show your calculation for at least one target distance below.

OPTIONAL — EXTRA CREDIT

A launcher sits at height H above the target line and fires at speed V_0 . Find the launch angle(s) θ needed to hit a target at horizontal distance D .



Dr. Cho's Extra Credit sketch — $H = 1.06 \text{ m}$, $V_0 = 5 \text{ m/s}$, $D = 1.5 \text{ m}$.

θ_1 (high arc) = _____° θ_2 (low arc) = _____°

Show your work below.
